W-A-450/9
JANUARY 3, 1992

FEDERAL SPECIFICATION

DETECTOR, MOTION, ULTRASONIC

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit and type. This specification covers Unit Number 1, Type IX, Ultrasonic Motion Detectors (transceiver, directional and semi-directional transducers, and omni-directional transmitter and receiver transducers).

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, of the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

3. REQUIREMENTS.

3.1 Qualification. The ultrasonic motion detector furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been tested and shall have passed the qualification tests specified in paragraph 4.1.

3.2 Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of operating within the temperature and humidity ranges indicated below:

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0 degrees C. to +49 degrees C. and relative humidity of 0% to 85%, non-condensing.

All units shall withstand the tests in paragraphs 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. All equipment supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 10 000 hours.

3.4 Operation. The ultrasonic motion detector shall consist of transmitter and/or receiver elements and necessary control circuitry to saturate the zone of detection with acoustical energy at frequencies above 16 kHz. Usable and published detection patterns shall be included in the manufacturer's literature. Failure to supply this information shall be cause for non-compliance of the sensor. Nominally the frequency ranges shall be between 19 kHz and 40 kHz. The detector shall, after initial setup, stabilize within five minutes after being powered on and after any alarm initiating stimulus is removed. A test target representative of a human that is no more than 1524 mm tall and about 410 mm wide, weighing about 36 kg, shall be used for walk testing, (see 4.1.1). The detector shall detect movement anywhere within the zone of detection before the test target has moved not more than 1500 mm at any velocity between 0.05 m/s and 3.05 m/s. Input voltage shall be 8 to 18 volts DC, reverse polarity protected, or 10 to 28 volts AC RMS, 60 cps. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment. Alarm and tamper switch impedance shall be as follows:

Open state:
- greater than 100,000 ohms.

Closed state:
- For solid state relays - less than 50 ohms.
- For mechanical relays - 0.1 ohms or less.

3.4.1 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature

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concerning performance data that would exceed the minimum requirements of this specification.

3.5 Ultrasonic assembly enclosure. The ultrasonic motion detector shall be housed in enclosures having a removable cover arranged to provide ready access for mounting, wiring, alignment, and for sensitivity and other internal adjustments. Mounting holes shall be provided in the back of the enclosure in positions that are readily accessible (sufficiently displaced from all internal parts and the enclosure side) when all internal parts and wiring are in place. No external hardware or fasteners except the cover fasteners shall be used. Accommodations for vibration resistant mounting blocks shall be provided.

3.6 Hooding. The detector shall go into the alarm state whenever any covering device is placed on or over the detector.

3.7 Cover. The cover shall be designed such that any attempted access to the interior shall cause a tamper condition. The tamper circuits shall be in the secure state whenever the cover is in place and shall go to their alarm state when the cover is moved sufficiently to reveal the inside or to introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.8 Terminals. Screw terminals shall be provided for interconnecting wiring. Access to the terminals shall require removal of the enclosure cover.

3.9 Switch and/or relay assembly. Alarm and tamper switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the published current rating shall not cause damage to the switch or contacts.

3.10 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this specification and shall be stated in the manufacturer's literature.

3.11 Maintainability. The ultrasonic motion detector shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the

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device or replaceable sub-assembly and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the detector beyond mere removal of the cover. The cover shall be designed in such a fashion that any attempted access to the interior shall cause a tamper condition. The tamper contacts shall be in an active state whenever the cover is in place and shall go into alarm state when the cover is moved sufficiently to see inside or introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.12 Life cycle. The ultrasonic motion detector alarm and tamper switches and/or relays shall be rated at not fewer than 1 000 000 operations under a load of 250 milliampere driven from a 48 volt DC source. Contact resistance shall not change more than +20% from the initial reading.

3.13 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Tests.

4.1.1 Performance test. The unit under test shall be mounted in accordance with the manufacturer's recommendations in a room large enough to accommodate the manufacturer's published zone of detection patterns. The room shall be free of all ultrasonic emitters. Doors and windows shall be tight fitting to avoid air movement which may contain noises in the ultrasonic frequency range. The unit's alarm contacts shall be connected to either an incandescent light bulb or light emitting diode. If an audio sounder is used, it should be placed in a baffle which absorbs ultrasonic frequencies to prevent interference with the sensor under test. The test engineer shall mark the zone of detection on the floor of the test area. From the point of origin of the detection pattern, radial lines shall be drawn to a point 3000 mm beyond the detector's zone of detection at 10 degree increments. The test shall be conducted in the following manner:

a. The engineer's arms shall be folded tightly across the chest and held fast to preclude extraneous movements that might affect the accuracy of detector response data. Instead of responding to a doppler shift created by a body mass, the detector shall measure the doppler shift response created by hand and arm movement.

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b. The engineer shall approach the detector straight on from the end of the radial.

c. The rate of movement shall be approximately 300 mm per second.

d. When detection occurs, the engineer shall stop walking and mark the location of the floor with a #1. The engineer shall reposition the arms and wait approximately 30 seconds for the detector to reset, and then begin walking toward the detector along the same radial until detection occurs again.

e. The walk test sequence shall be repeated until the entire length of the radial line is tested. All radial lines shall be tested in the same manner.

f. Repeat the tests marking the locations with a #2. Note the differences. If the two marks for each location exceeds 300 mm, the test shall be performed a third time.

g. Use the marks placed on the floor to determine the actual zone of detection pattern. Compare the actual pattern to the projected pattern in the manufacturer's published literature. The actual zone of detection pattern shall be within plus or minus 5% of the manufacturer's claims.

4.1.2 Stability test. The unit submitted for qualification testing shall perform its intended function in a stable environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

4.1.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.4. Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0 degrees C. The unit shall then be tested as specified in

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paragraph 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. Units shall be tested for compliance with 3.13 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.

4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil agency marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

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6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration
Federal Supply Service, Furniture Commodity Center
Washington, DC. 20406

Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Alarms.

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6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.2 Invalid alarm. Any other alarm.

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